# VISHNU PULIPAKA

Machine Learning Engineer

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## Summary


Applied researcher focused on large language models, information retrieval, and systems-level inference optimization. Experience building RAG and hybrid retrieval pipelines, evaluating vector and BM25 indexes, and optimizing ML inference (PyTorch → ONNX → TensorRT) and distributed data pipelines for production. Currently investigating C++ and PyTorch inference runtimes for reward-model scoring and federated learning; two research papers on LLM fairness and multilingual training are under review.


## Work

### Software Engineer

Jun 2022 – Jul 2024

*4SS Software Solutions* • India

- Integrated LLM-powered capabilities into Python and Java services, supporting prompt execution, evaluation workflows, RAG pipelines, and production AI applications.
- Developed REST APIs and data pipelines for model inference, experiment tracking, retrieval workflows, and downstream application integration.
- Built Kafka-based asynchronous services for scalable AI workload processing and distributed replication, reducing interservice processing time by 22%.
- Applied Redis caching and backend optimizations to model-enabled services, reducing average API response latency by 25%.
- Optimized SQL queries and inference-supporting services to improve reliability, throughput, and production performance.
- Established GitHub Actions pipelines for testing and deploying ML-enabled services to AWS, reducing deployment time by 20%.

## Project

*RAG Evaluation Platform with CI/CD Quality Gates*

- Achieved 100% retrieval hit rate and 92% faithfulness, with CI gates blocking retrieval, citation, and generation regressions.
- Built a RAG evaluation platform comparing BM25, vector, and hybrid retrieval across three LLM providers and seven metrics.

*TensorRT Inference Optimization Engine*

- Reduced P99 inference latency from 21.4 ms to 4.9 ms while maintaining 95.03% accuracy across 10,000 images.
- Converted ResNet-50 from PyTorch through ONNX to FP16 TensorRT, increasing throughput from 97 to 347 FPS.

*Distributed CDC Pipeline with Exactly-Once Delivery*

- Validated zero data loss under TaskManager failures and network partitions.
- Built a CDC pipeline handling 50,000 writes per minute with exactly-once processing.

## Volunteer

### Data Engineer (Volunteer)

Mar 2026 – Present

*Saayam For All*

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- Hardened 4 AWS Lambda workflows using validation, exception handling, and reliable PostgreSQL connection management.
- Migrated mock CSV inputs to PostgreSQL RDS APIs delivering eight dashboard metrics with automated payload validation.

## Education

### Montclair State University

Sep 2024 – Jan 2026

**Data Science** | Master of Science

GPA: 3.87/4.0

## Skills

**Programming**
Python, SQL, C++

**Machine Learning & NLP**
PyTorch, Hugging Face, Transformers, NLP, LLMs, RAG, Embeddings, Model Evaluation, Federated Learning

**Information Retrieval**
Vector Search, Hybrid Search, BM25, Semantic Search, Inverted Indexes, Retrieval Evaluation, Reranking

**MLOps & Inference**
TensorRT, CUDA, ONNX, MLflow, Optuna, Evidently, GitHub Actions

**Systems & Backend**
FastAPI, Flask, PostgreSQL, Redis, Apache Kafka, PySpark, Docker, Kubernetes, AWS, Linux, Networking

**Personal Projects**
RAG Evaluation Platform with CI/CD Quality Gates, Sentence Transformers, TensorRT Inference Optimization Engine,

Distributed CDC Pipeline with Exactly-Once Delivery, PyFlink, Apache Iceberg

**Computer Science Fundamentals**
Data Structures, Algorithms, Indexes